IS601 | DATABASE ASSIGNMENT

FastAPI + PostgreSQL

Docker Compose, pgAdmin, and SQL

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GitHub repository

https://github.com/tkg1121/fastapi-postgres-calculator

Successful GitHub Actions workflow

https://github.com/tkg1121/fastapi-postgres-calculator/actions/runs/41763079480

This project runs the FastAPI calculator, PostgreSQL, and pgAdmin as three Docker Compose services. The screenshots document each SQL command, its output, and the final database records.

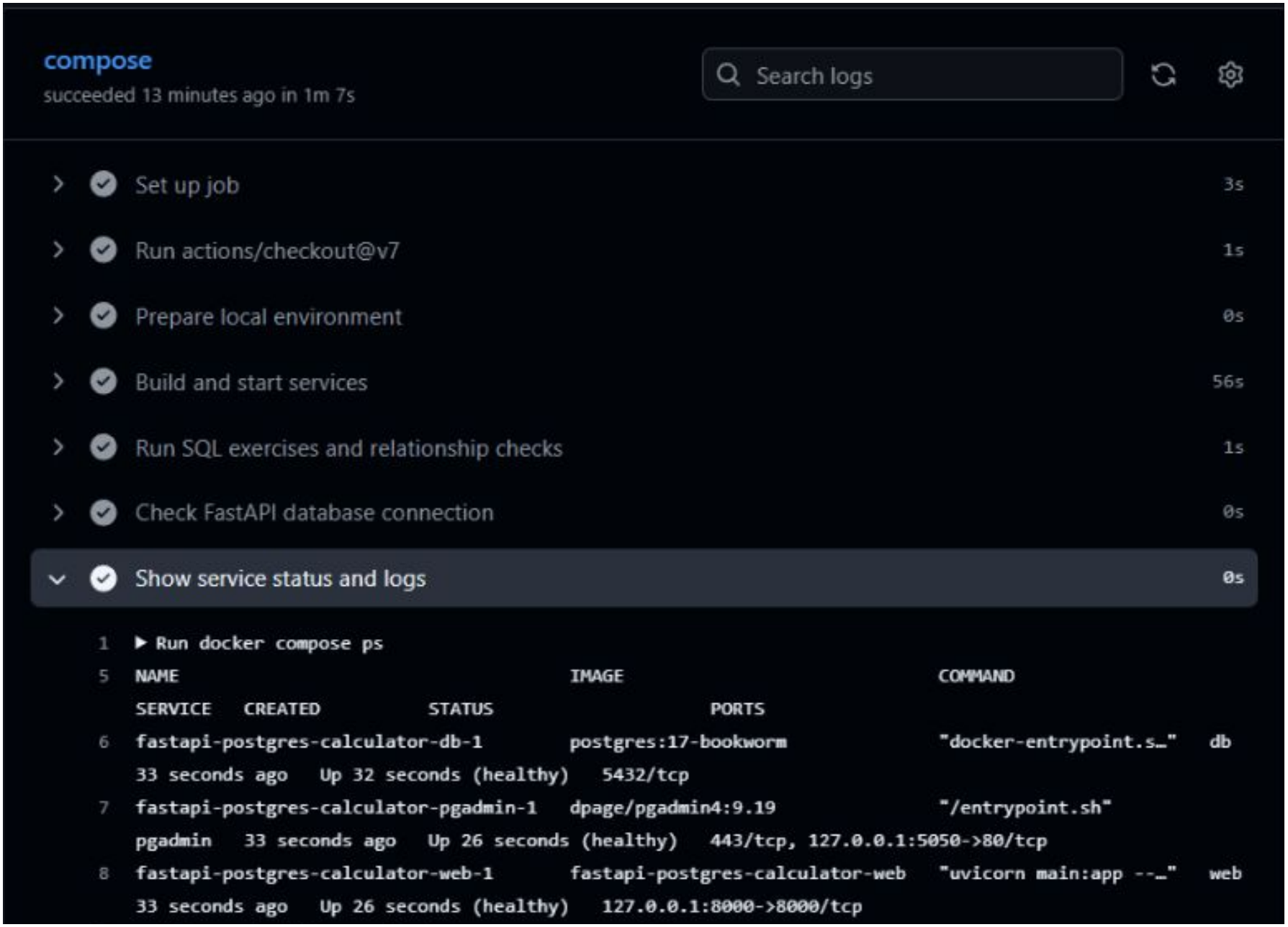
The users and calculations tables have a one-to-many relationship. The calculations.user_id foreign key references users.id. ON DELETE CASCADE removes a user's calculations when that user is deleted.

After the required update and deletion, calculation 1 belongs to Alice with a stored result of 6, and calculation 3 belongs to Bob with a result of 20. The changed result is the assignment's SQL update example, not a new arithmetic result for 2 + 3.

Validation: 130 application tests passed. The Compose workflow also verified the SQL exercises, rejection of invalid foreign keys and duplicate user fields, cascade behavior, and the application's connection to fastapi_db. Relationship checks use a transaction and roll back their record changes.

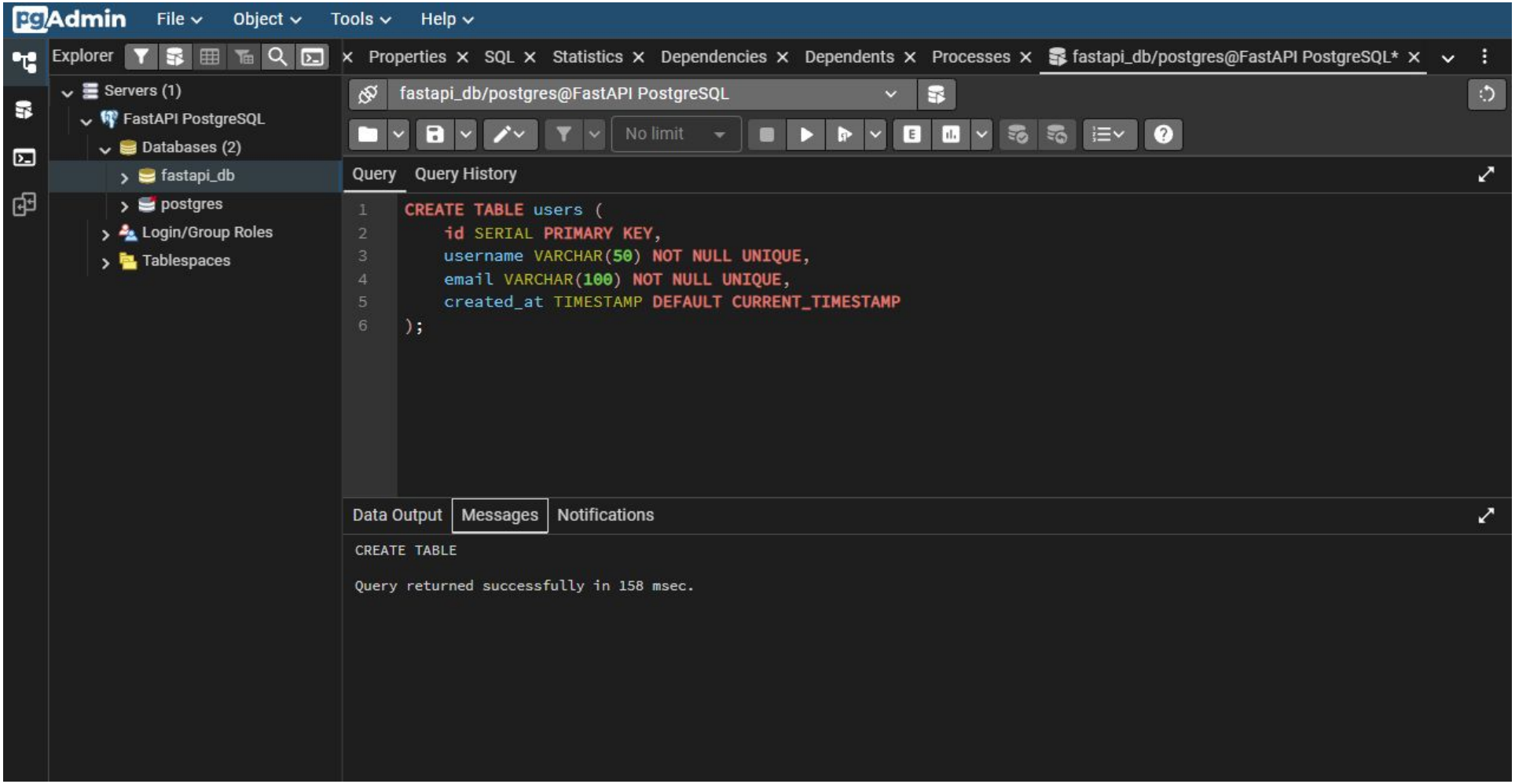
Docker Compose services in GitHub Actions

The GitHub Actions Compose job shows FastAPI, PostgreSQL, and pgAdmin running and healthy. These service logs are from the CI runner.



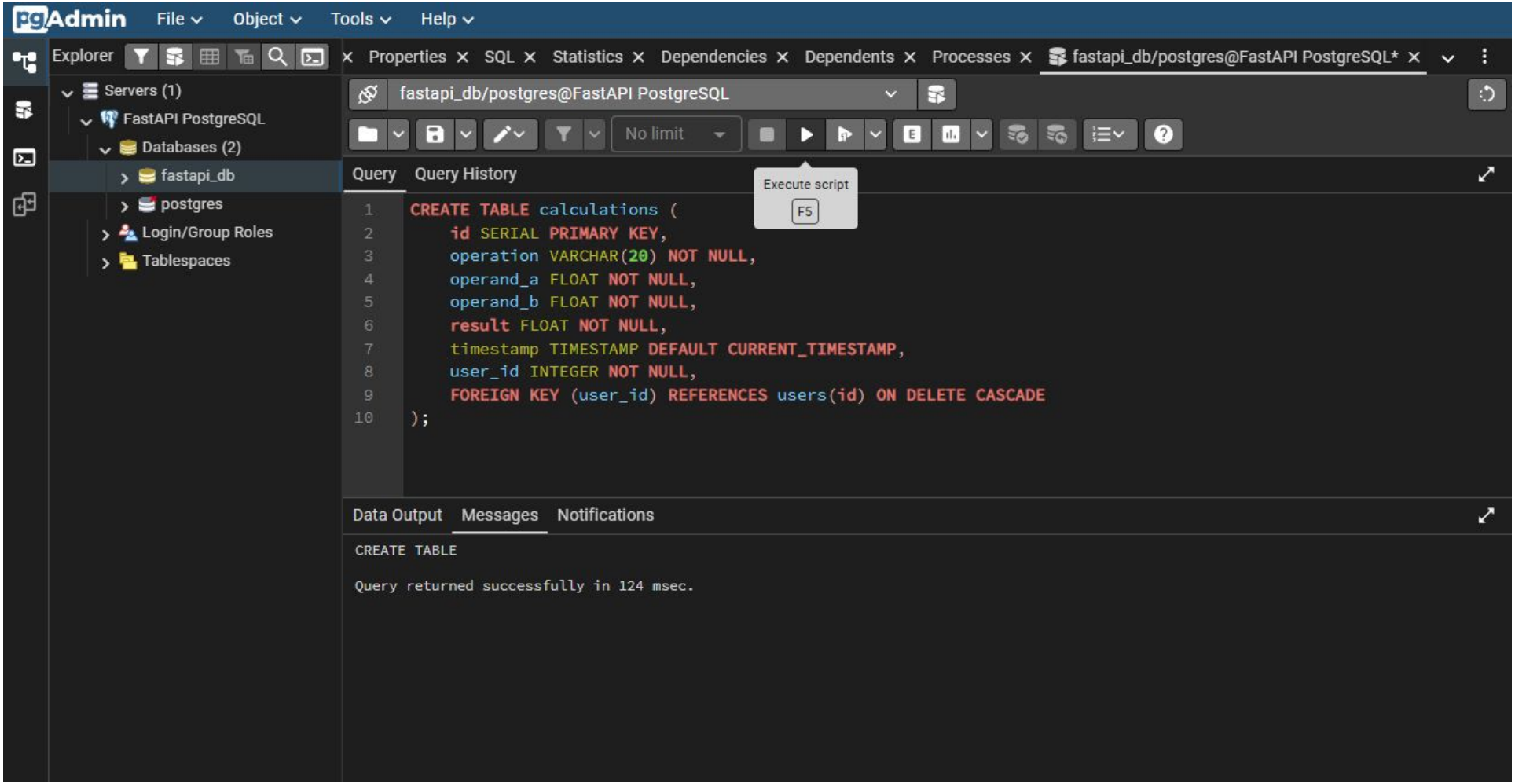
Create the users table

Creates a primary key, unique usernames and email addresses, and a default creation timestamp.



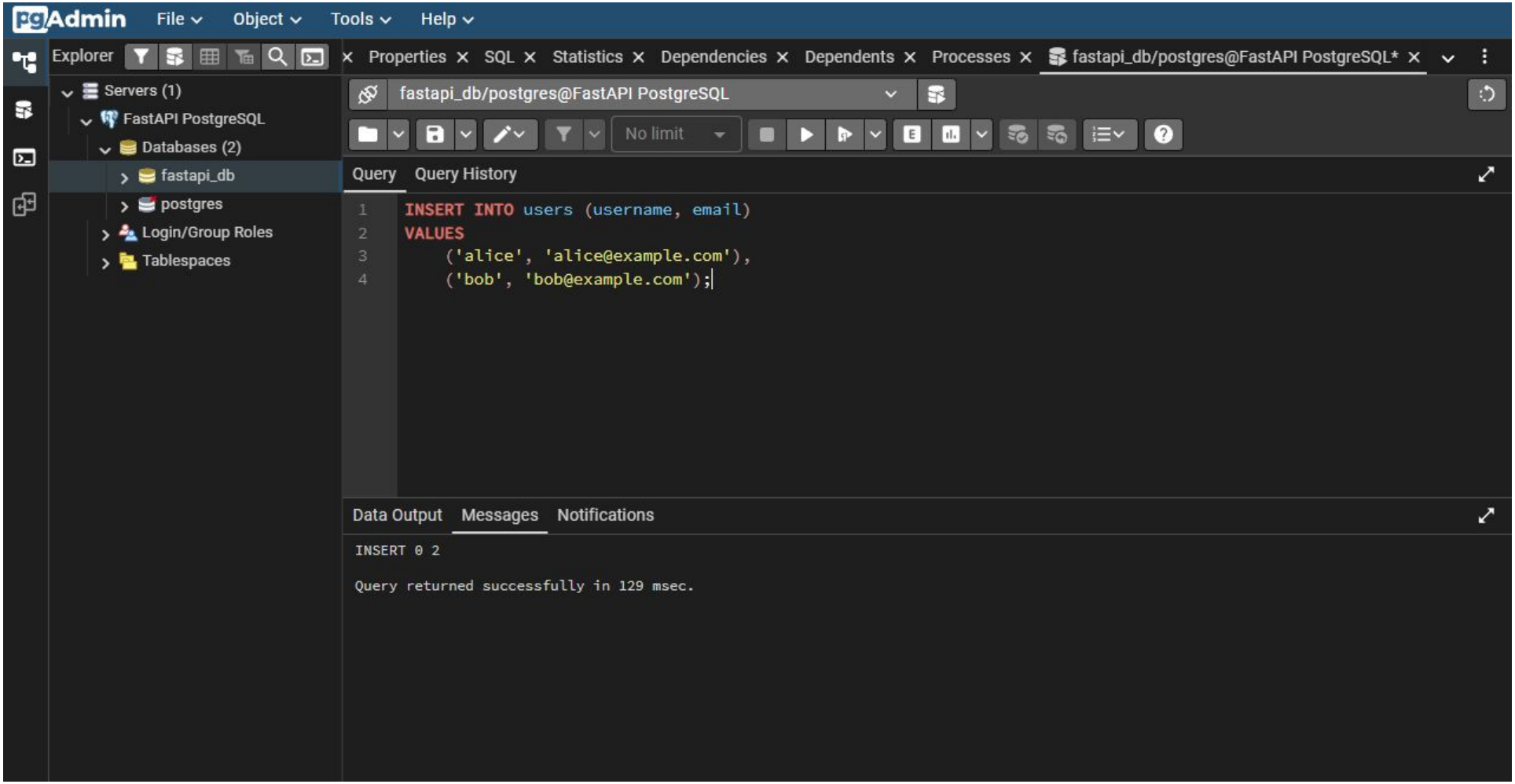
Create the calculations table

Creates the calculation records and a foreign key to users, with ON DELETE CASCADE.



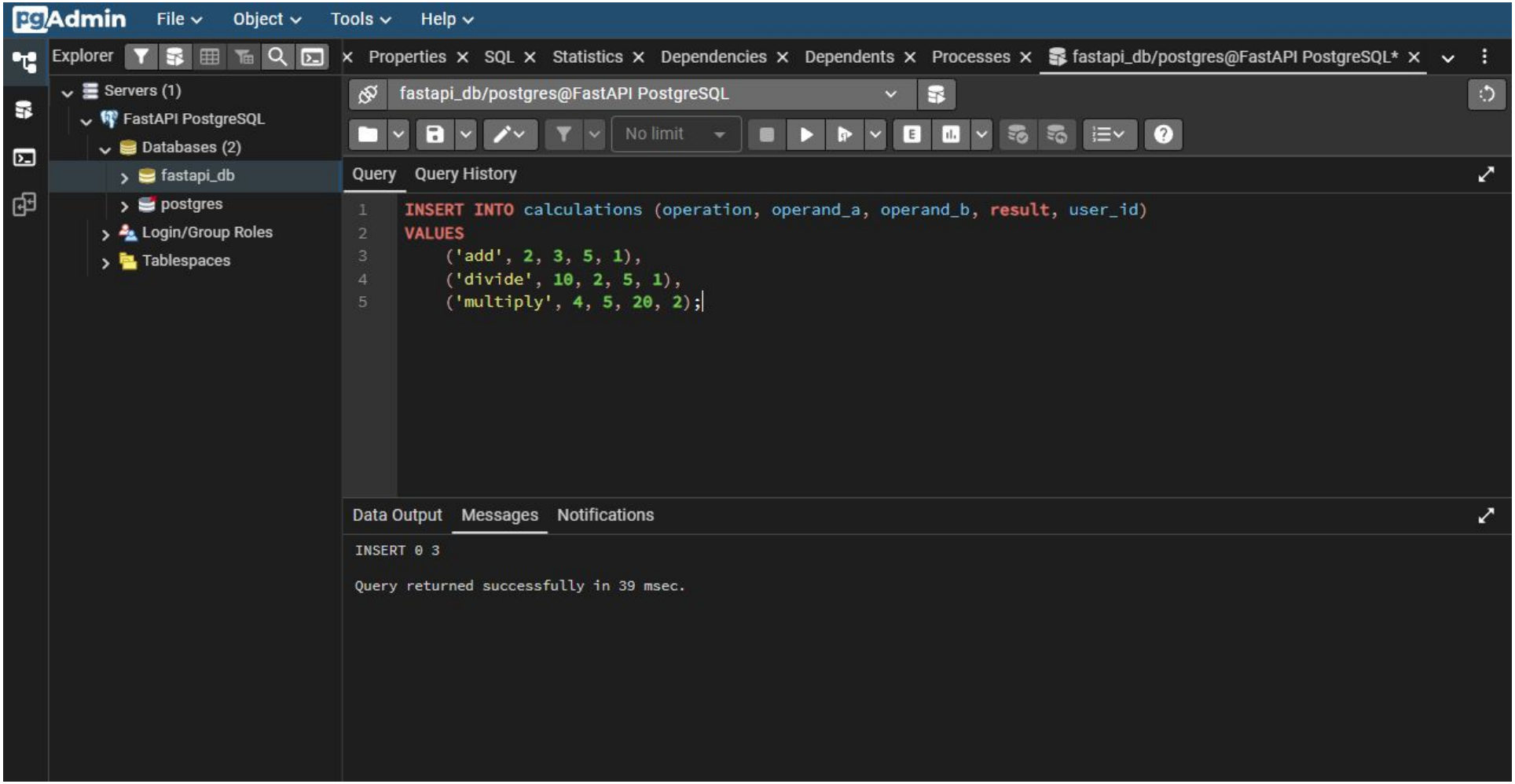
Insert two users

Adds Alice and Bob. Their generated IDs connect the subsequent calculations to their owners.



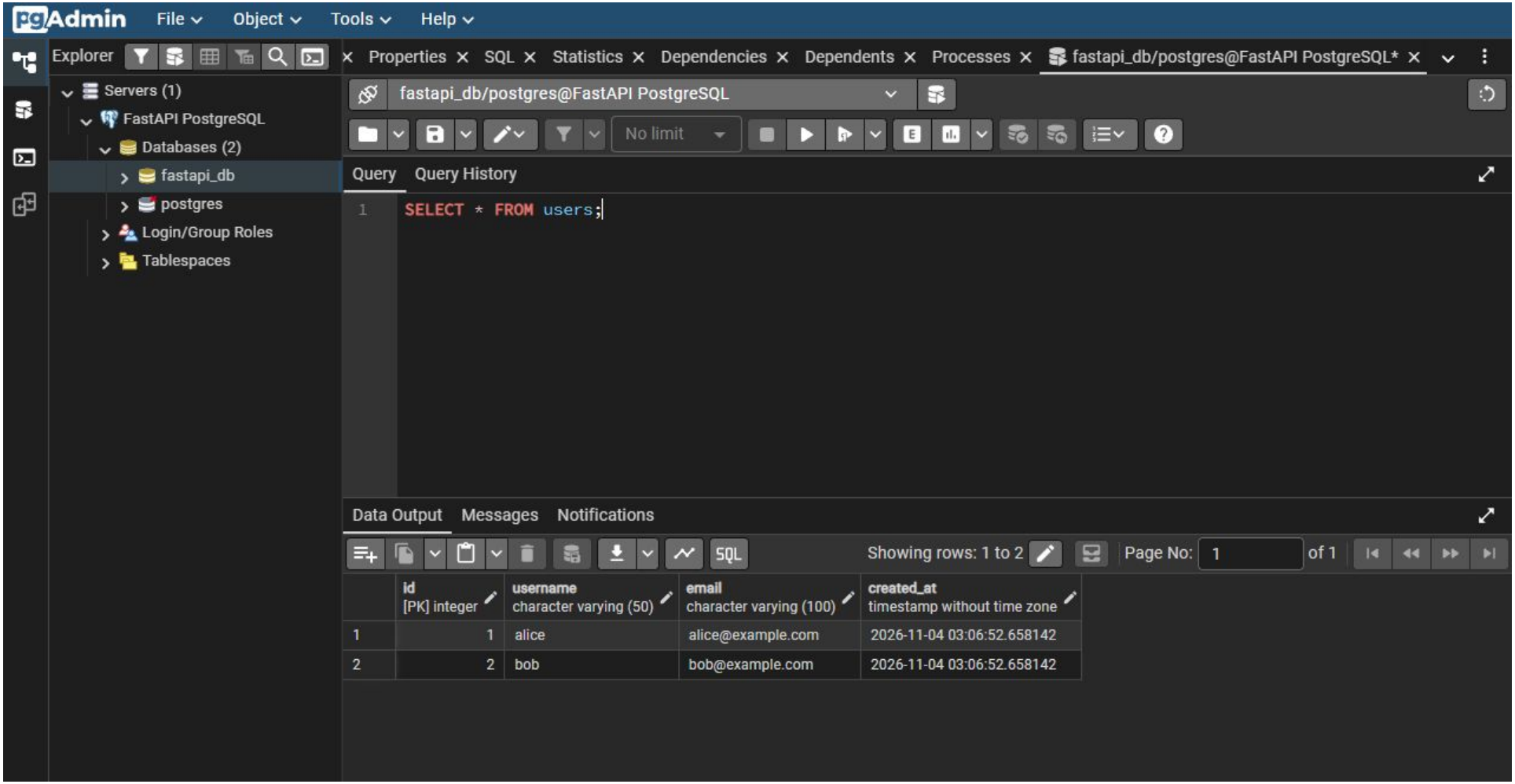
Insert three calculations

Adds Alice's addition and division records and Bob's multiplication record.



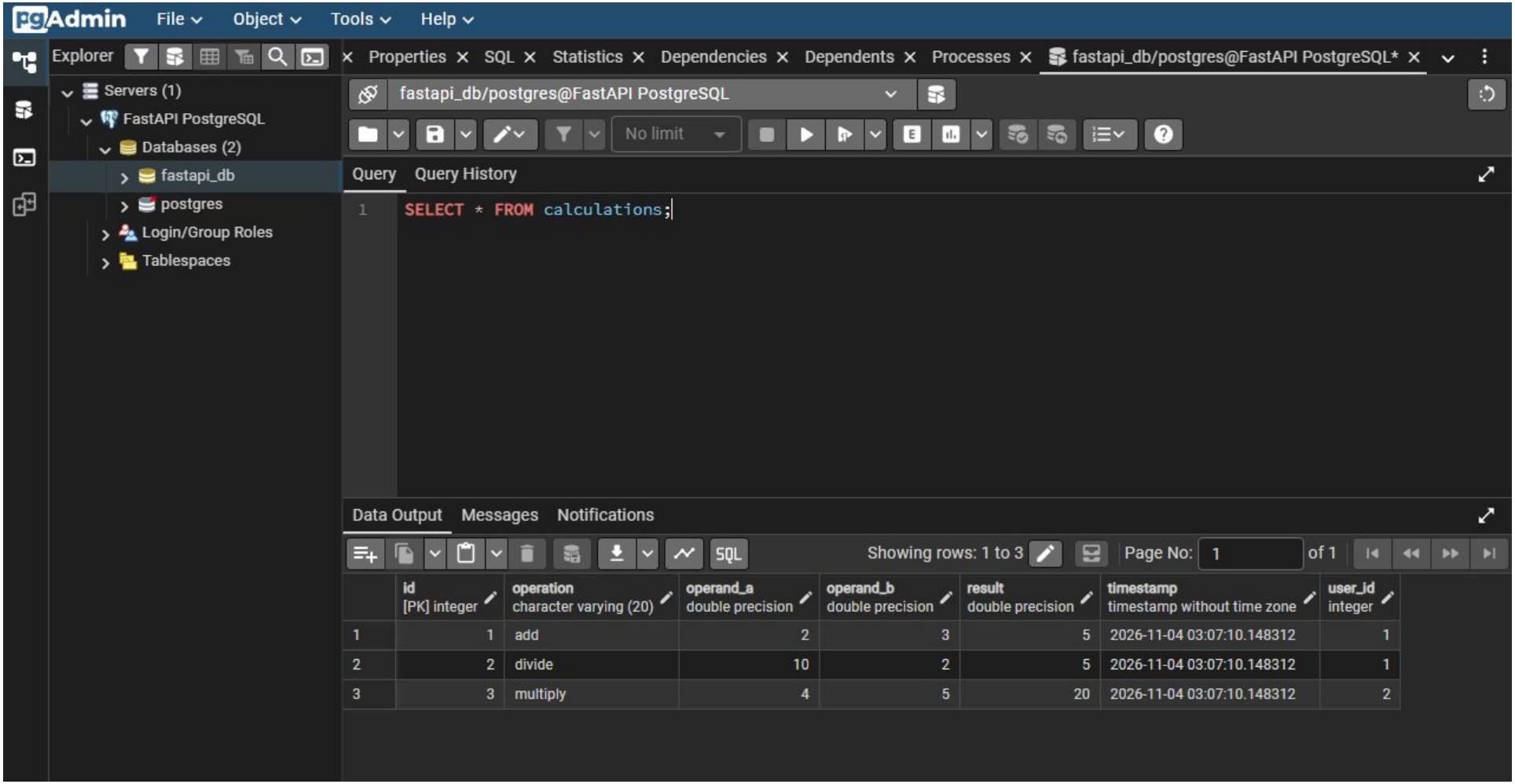
Query the users

SELECT * FROM users returns the two inserted user records.



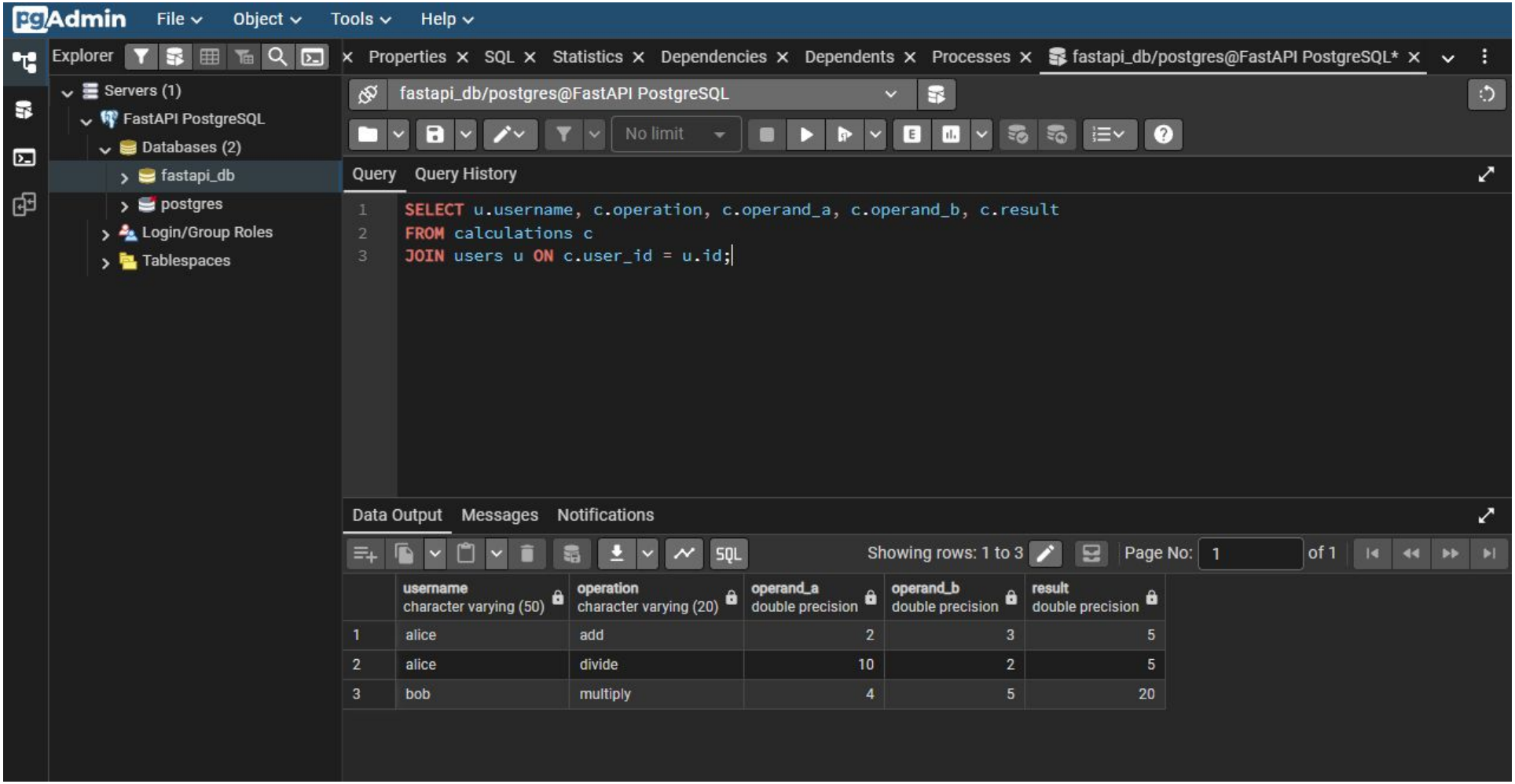
Query the calculations

SELECT * FROM calculations returns all three records before the update and deletion.



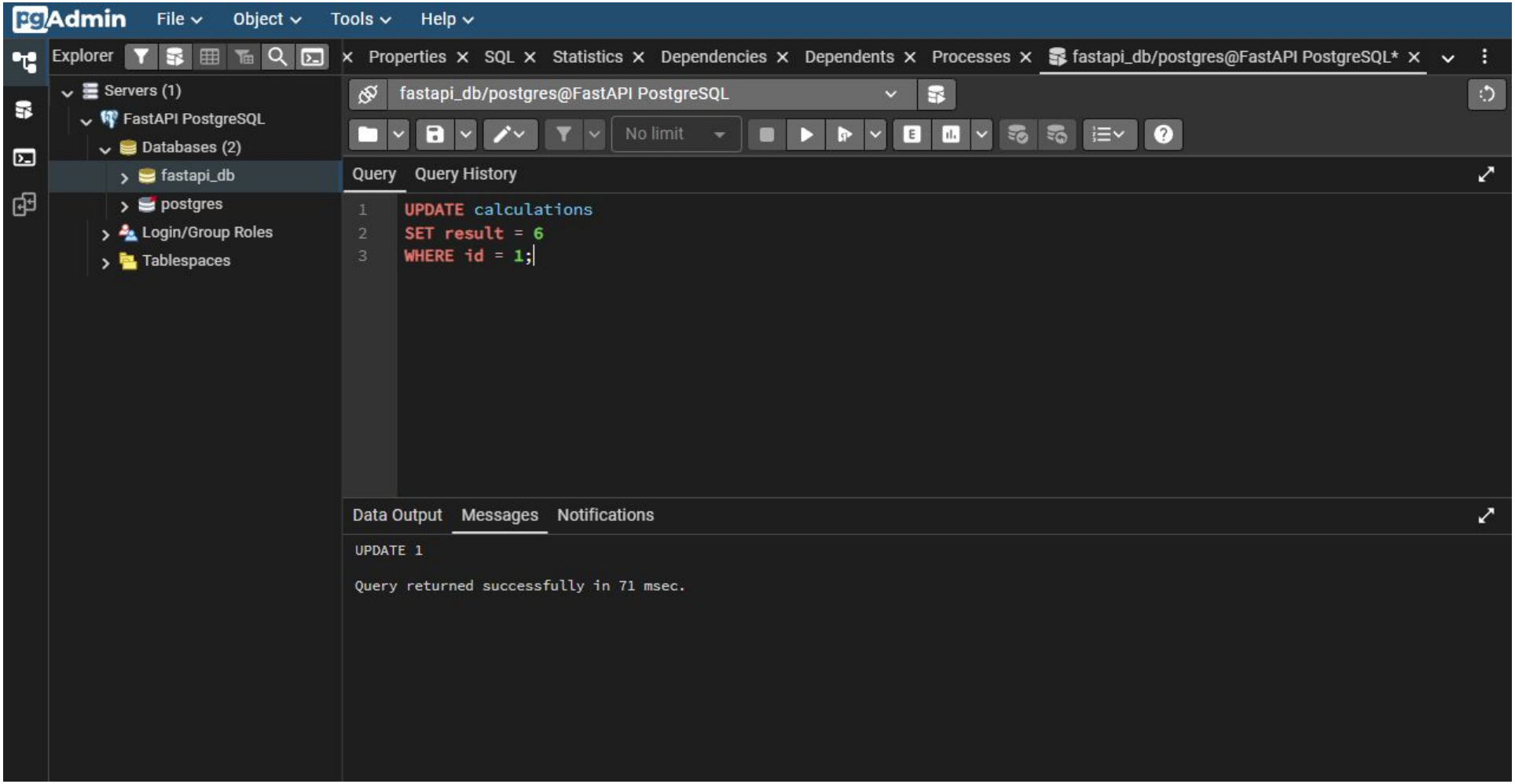
Join users and calculations

The join matches calculations.user_id to users.id and displays each operation with its owner's username.



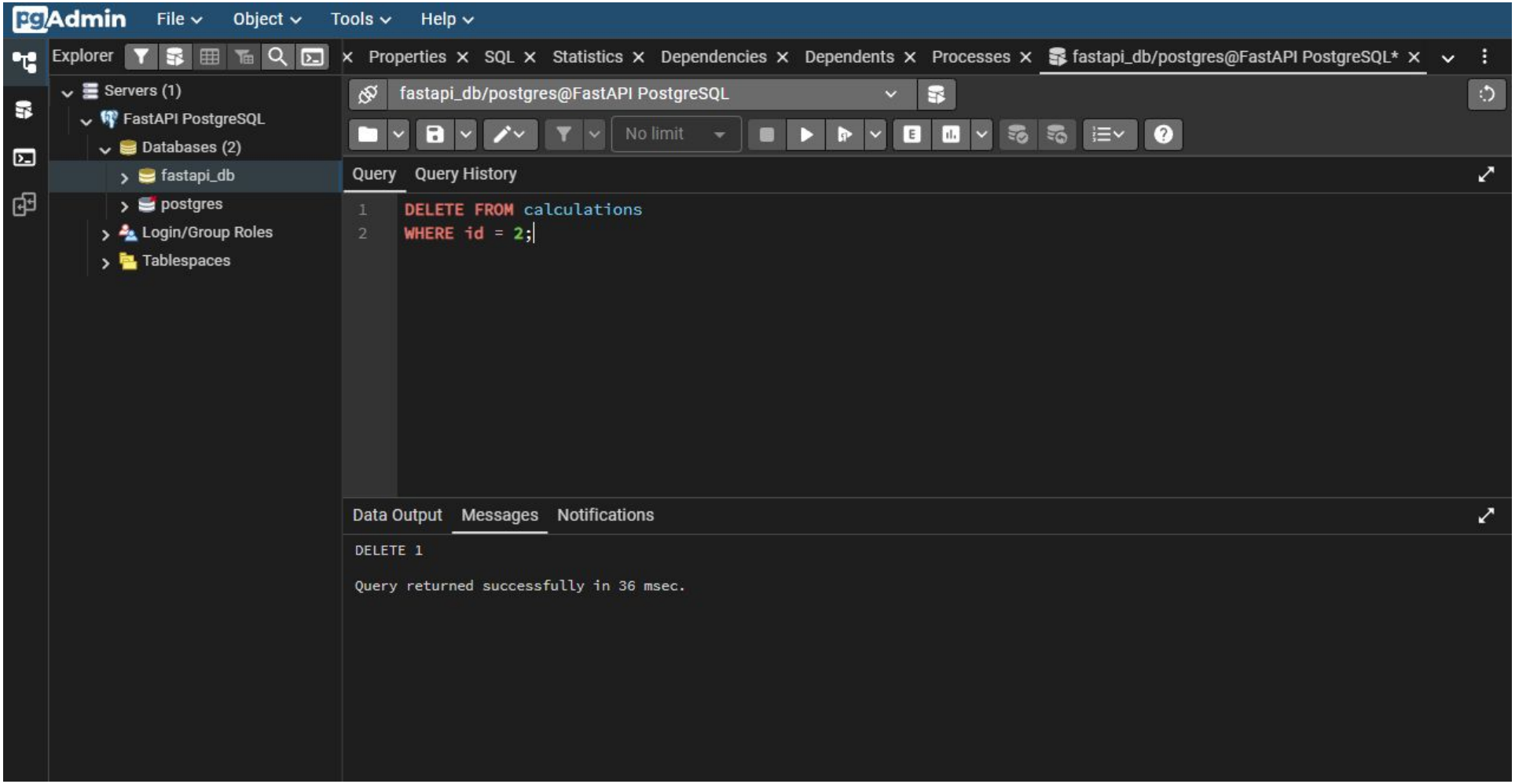
Update one calculation

Changes the stored result for calculation 1 to 6. This is the requested SQL update demonstration; 2 + 3 still equals 5.



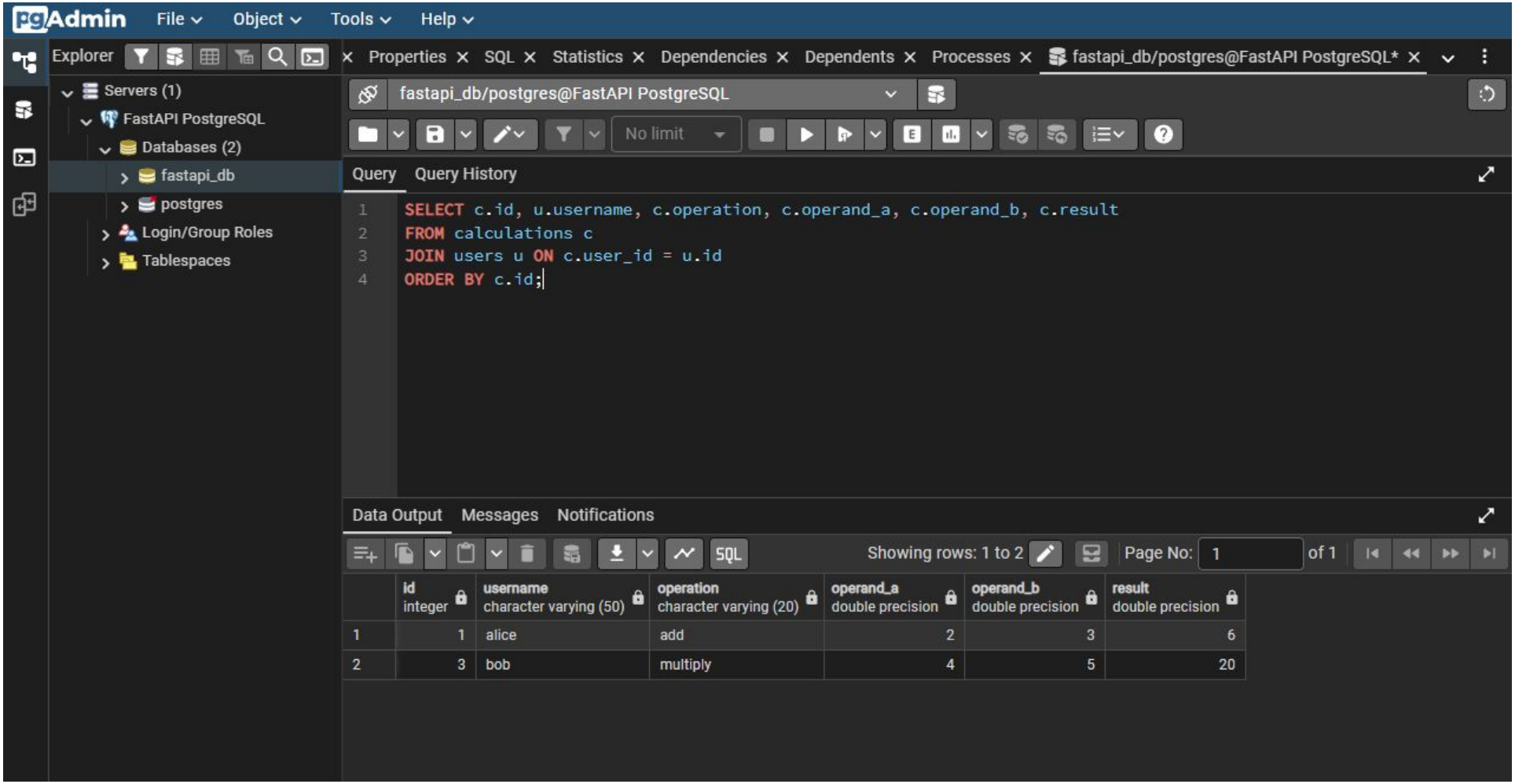
Delete one calculation

Deletes calculation 2 using its primary key. The WHERE clause limits the deletion to that record.



Verify the final records

The remaining records are calculation 1 (Alice, add, result 6) and calculation 3 (Bob, multiply, result 20).



Automated verification

GitHub Actions passed the application tests and the Compose job, including SQL scripts 01-11 and the FastAPI database connection check.

